	Registration (live → fixed)	Host / nucleus segmentation	TUNEL call	Parasite localisation	Measurement & statistics
Published	by hand	hand-drawn ROIs	by inspection	15 ROIs + 3 backgrounds each, by hand	ROI Manager → Excel; Mann–Whitney
Codex CLI	SIFT + RANSAC, ECC refined (1.6 px)	T1 DAPI threshold; T2 Cellpose cyto3	largest-gap cutoff	red-channel peaks + watershed: 10 kept	annulus background; no test
Exploratory Agentic-J	TurboReg rigid on brightfield	T1 StarDist; T2 Cellpose cyto3 (7 hosts)	largest-gap cutoff	C2 + C3 spot evidence: 22 kept	annulus background; no test
Conventional Agentic-J	TurboReg / StackReg rigid	T1 DoG + watershed; T2 Cellpose cyto3 (6 hosts)	2-component GMM	micro-sam vit_b_lm: 105 → 5 kept	annulus background; no test
Conventional Agentic-J + human	lab ROIs given, origin translated	lab ROIs (given)	largest-gap cutoff	lab ROIs paired 1:1 with backgrounds	native 16-bit; Mann–Whitney

Manual Human feedback Automatic (Agentic-J) Automatic (other agent)